Dylan Sinnott

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Education

University College Dublin – UCD Michael Smurfit Graduate Business School **August 2026 – Present**
MSc Quantitative Finance (Expected August 2027) *Dublin, Ireland*

University of Galway **September 2022 – May 2026**
BSc Applied Mathematics (First Class Honours) *Galway, Ireland*

Relevant Coursework

Mathematical Methods I (100%), Linear Algebra (97%), Mathematical Methods II (95%), Non-Linear Elasticity (95%), Mathematical Modelling (90%), PDEs, Numerical Analysis

Projects

Benchmarking Physics-Informed Methods for Black-Scholes Option Pricing **May 2026 – July 2026**
Python · NumPy · SciPy · Pandas · Matplotlib · PyTorch · PyKAN · Git

- Implemented analytical Black-Scholes pricing, a Crank-Nicolson finite difference solver, and PINN/PIKAN models for European call options.
- Benchmarked pricing accuracy, runtime and volatility sensitivity; Crank-Nicolson achieved MSE 9.6×10^{-5} in 0.29 seconds.
- Selected to speak at PyCon Ireland 2026 on benchmarking classical numerical and physics-informed machine learning methods for option pricing.

Absolute Continuity and the Fundamental Theorem of Calculus **September 2025 – April 2026**
Undergraduate Dissertation *University of Galway, Ireland*

- Authored a 64-page dissertation on the Fundamental Theorem of Calculus for absolutely continuous functions, developing the supporting theory of Lebesgue integration and bounded variation.

Experience & Leadership

Irish Student Managed Fund (ISMF) **October 2026 – Present**
Director, Quantitative Finance *Dublin, Ireland*

- Developing a 45-minute training lecture for quantitative trading analysts on systematic position sizing and risk management, covering volatility targeting, exposure limits and the Kelly criterion.

UCD Student Managed Fund **September 2026 – Present**
Equity Analyst *Dublin, Ireland*

- Research Aker BP and Vår Energi, analysing financial performance, production trends and energy market developments to prepare company reports and support investment discussions.

University of Galway **July 2025 – August 2025**
Undergraduate Research Assistant *Galway, Ireland*

- Awarded an Aerosol Society undergraduate research bursary; designed and conducted over 100 indoor air quality experiments during a six-week research project.
- Authored a technical report and received an Early Career Travel Award to present findings at the Annual Aerosol Conference in York.

Technical Skills

Programming Languages: Python
Python Libraries: NumPy, SciPy, pandas, Matplotlib, PyTorch
Mathematics: Numerical Analysis, Partial Differential Equations, Linear Algebra, Optimisation, Probability
Tools: Git, GitHub, VS Code, Microsoft Excel, Bloomberg, LaTeX

Awards and Certifications

University Scholar — University of Galway (2025)
Bloomberg Market Concepts (2026)

Interests

Chess: Top 2% of rapid players on Chess.com.
Running: Completed the Donegal Half Marathon (2025).
Fantasy Premier League: Best overall finish: 1,744th globally.
Creative Writing: Published short fiction in university publications.